

Layer Classification — Token Lexicons, Confidence & LLM Override

Input: ModuleRecord or FunctionRecord
(name + source_object)

STEP 1: Tokenize

- Split name on `_` and camelCase boundaries
- Split source_object on `/` and `\\`
- Lowercase all tokens
- Remove numeric suffixes and single chars

STEP 2: Match tokens against Layer Lexicons (9 built-in layers)

 Application app, main, task, user, cli, logic, business, service, workflow	 GUI gui, widget, screen, display, ui, view, button, dialog, panel	 Communication comm, can, spi, uart, i2c, eth, tcp, udp, modbus, iin, protocol	 Driver drv, driver, hal, gpio, adc, dac, pwm, timer, flash, wdg, rtc	 System sys, system, init, startup, config, clock, reset, power, pwr, sleep	 Security crypto, aes, sha, rsa, ssl, tls, auth, key, cert, secure, hash	 Middleware rtos, os, freertos, cmsis, thread, queue, mutex, semaphore, event, heap	 Runtime crt, libc, runtime, math, stdlib, newlib, startup, vectors, exception
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Lexicon Score per layer: $score_L = \frac{matching_tokens}{total_tokens} \times boost_factor$
boost_factor: 1.5 if source_object path matches layer pattern, else 1.0

STEP 3: Compute Classification Confidence

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best_layer = argmax(score_L for all L)
second_best = second highest score

confidence = best_score - second_best_score

Thresholds:
confidence >= 0.6 -> classification = best_layer
confidence 0.3-0.59 -> classification = best_layer (low_confidence flag)
confidence < 0.3 -> classification = None (ambiguous)
```

ambiguous?

STEP 4: LLM Override (optional, AI-assisted)

```
Triggered when:
• confidence < 0.3 (ambiguous), OR
• multiple layers within 0.1 of each other

LLM prompt: "Classify {name} from {path} into one of: [layers]"

Acceptance:
• LLM response must match a valid layer name
• Sets confidence = 0.5 + 0.1 (fixed bump)
• Marks ai_override = 'llm_classification'
• If LLM unavailable -> remains None
```

Output: item.layer = str | None, item.layer_confidence = float